

PSF-Zero

A compiler engine that resolves quantum-circuit compilation bottlenecks with closed-form algebra

The problem

Running a circuit on a quantum computer requires a compilation (transpilation) step beforehand. Leading existing tools (e.g. Qiskit) are known to suffer an unexpected 10x–several-hundred-x slowdown in compile time under a condition that is common in real deployments: when qubit placement on the hardware is nearly or fully saturated. This directly affects cloud quantum-computing turnaround time, cost, and the pace of R&D iteration.

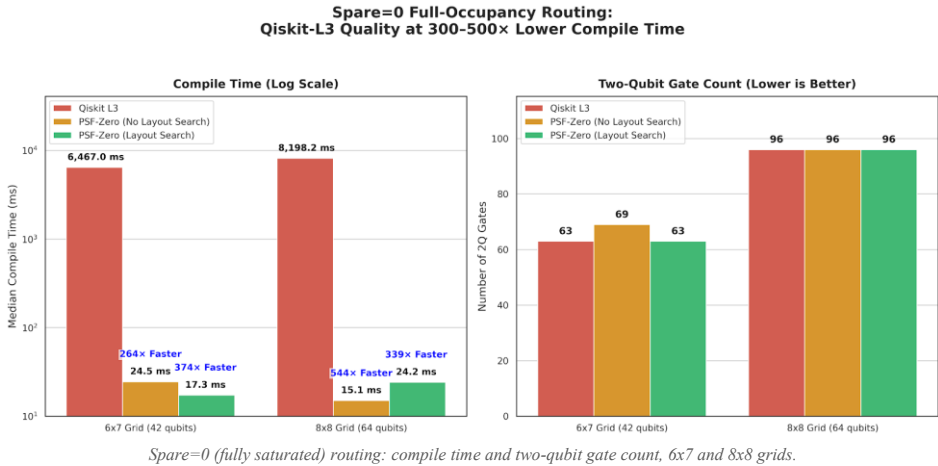
PSF-Zero's approach

PSF-Zero replaces Qiskit's own iterative re-optimization of two-qubit gates with a single closed-form (Cartan/KAK) computation, so the synthesis step other tools spend repeated search cycles on is done once, deterministically. Because gate synthesis no longer depends on Qiskit's `optimization_level`, the existing toolchain's own qubit-routing stage can run under a much lighter configuration than it otherwise would, without sacrificing correctness. It is this decoupling — not a novel routing algorithm — that sidesteps the well-documented slowdown existing routers hit when hardware is nearly or fully occupied, delivering compilation hundreds of times faster.

Measured results

Comparison	Condition	Result
Qiskit (optimization_level=3)	General dense circuits (15–1000 qubits)	2.5–5x faster
TKET (FullPeepholeOptimise)	Same circuit set (10–160 qubits)	150–270x faster
Qiskit (optimization_level=3)	Worst case: coupling map nearly/fully saturated (42–64 qubits)	279–544x faster — measured
Real hardware (IBM Quantum)	15 qubits, 10 jobs	Fidelity statistically on par with Qiskit default; compile time 14–16x faster
Output reproducibility	300 trials, identical input	Always returns the identical circuit (zero variance)

* Output gate-count parity is measured under a configuration chosen to match the target hardware's native two-qubit basis. Full measurement conditions are available on request.



Our approach to verification

Every number in this document is backed by raw data in a public repository and a reproducible benchmark script. Predictions are pre-registered before each measurement is taken, and any error found is kept as a public, dated correction rather than silently replaced. We prioritize verifiability over the kind of overstated performance claims common in the quantum-computing field.

Proposal: Proof of Concept (POC)

Using your organization's actual quantum circuits (or synthetic circuits close to your use case), we would compare:

- Compile (transpile) time
- Output circuit gate count and depth
- Real-hardware fidelity (where feasible)

Estimated duration: a few days to a few weeks. Can be run in your environment or in an environment we provide.